CAP 6908 DIS (Fall 2026)

When does a LiDAR detector stop relying on the object?

A stratified study of evidence dependence in 3D detection confidence

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Week of Oct 19. Cross-architecture comparison. The same deletion protocol is repeated on one one-stage detector and one two-stage detector. Deliverable: a cross-detector comparison table.

Progress

PointPillars remains the one-stage arm, and that run was not repeated. PV-RCNN is the two-stage arm. The checkpoint is the public OpenPCDet release `pv_rcnn_8369.pth`, loaded as published, with no training.

Before any edit, the checkpoint has to reproduce the published KITTI validation Car result. Official moderate 3D average precision at 11 recall positions is listed as 83.61. This run scores 83.6974. The easy, moderate, and hard values are 89.3533, 83.6974, and 78.7166. Average precision at 40 recall positions is stored with the run and is not the published comparison.

Eligibility is recomputed from PV-RCNN's own Car predictions: score at least 0.7 and at least one point inside the predicted box. That set contains 16,275 cars. PointPillars had 11,125 under the same rule. The two sets are not paired by frame or prediction index, and each retained score is divided only by the original score from the same detector.

The bins, fractions, seed, and three edits are the ones locked in weeks 6 and 7. Interior removal deletes points inside the predicted box, the placebo deletes a same-size empty region, and context removal deletes points outside the box. Retained confidence is the new score divided by the original score when the returned box overlaps the original at 3D IoU 0.5, and 0 otherwise. A miss stays in the mean, and score survival also requires the returned score to be at least 0.5.

PV-RCNN predictions and every edited forward use batch size 2. Fraction 0 is the saved unchanged prediction. A live probe in each range bin returned the stored box at 3D overlap at least 0.99 and moved its score by less than 0.01. The finished run has 244,125 result rows and no correctness failures. The checkpoint, the prediction file, and the 7,481 Velodyne files are unchanged.

Findings

Table 1 is complete interior removal. Both detectors keep the same range and occupancy ordering, while their magnitudes differ.

For PointPillars, retained confidence falls from 0.193 within 20 m to 0.074 beyond 40 m. For PV-RCNN it falls from 0.228 to 0.038. The sparse bins are lower than the dense bins for both checkpoints: 0.086 against 0.190 for PointPillars, and 0.021 against 0.215 for PV-RCNN. A box with fewer interior points loses more of its score when those points are removed.

Predicted volume does not produce that ordering. The small, middle, and large bins retain 0.177, 0.154, and 0.178 for PointPillars. The same bins retain 0.149, 0.127, and 0.151 for PV-RCNN. The middle bin is slightly lower for both, and the small and large bins sit together.

Table 1: Complete interior removal. Retained confidence stores a miss as zero. Score is the share still at 0.5 or above, and IoU is the share still overlapping at 0.5. Each n is that detector's own eligible set.

	PointPillars (one-stage)				PV-RCNN (two-stage)			
Stratum	n	Retained	Score	IoU	n	Retained	Score	IoU
Sensor range								
0–20 m	4,864	0.193	13.5%	45.5%	5,496	0.228	23.8%	38.0%
20–40 m	4,868	0.149	10.3%	34.7%	7,110	0.103	10.3%	18.7%
40 m and farther	1,393	0.074	3.8%	18.1%	3,669	0.038	3.3%	7.4%
Interior points								
0–19 points	370	0.086	4.3%	19.2%	2,371	0.021	1.6%	4.6%
20–99 points	3,419	0.098	5.8%	24.5%	5,668	0.054	4.9%	10.8%
100 or more	7,336	0.190	13.6%	44.2%	8,236	0.215	22.5%	36.0%
Predicted volume								
Under 8 m^3	1,299	0.177	12.5%	41.3%	1,360	0.149	15.3%	25.7%
8–12 m^3	9,082	0.154	10.4%	36.8%	13,460	0.127	12.8%	22.2%
12 m^3 or larger	744	0.178	13.8%	37.4%	1,455	0.151	15.8%	23.9%

In every interior cell, a box comes back more often than a score of at least 0.5. Table 2 is the same fraction-1 cut for the two controls. Neither copies the interior drop: the empty-region means stay between 0.997 and 1.000, and context removal stays between 0.986 and 1.046.

Table 2: Fraction-1 retained confidence for the two controls, by sensor range. The denominators are the eligible counts in Table 1.

Range	Empty region		Context	
	PointPillars	PV-RCNN	PointPillars	PV-RCNN
0–20 m	1.000	1.000	1.018	0.986
20–40 m	1.000	0.998	1.045	0.991
40 m and farther	0.999	0.997	1.046	0.990

The empty-region comparison is still limited by point count. At fraction 1, PV-RCNN removes about 1,185 interior points from a nearby car and about 330 points from the matched empty region. Beyond 40 m those averages are about 25 interior points and about 4 empty-region points. The control rules out deletion of an arbitrary same-sized patch. It does not rule out a deletion matched on the number of points.

The numerical gap between the checkpoints is not an isolated effect of architecture. PointPillars and PV-RCNN differ in training, operating score, and the cars each one emits. The shared result is the ordering: longer range and fewer interior points leave less confidence after the interior is emptied, and predicted volume does not.

Next steps

The week of Oct 26 is the first complete draft of the CVPR report, due Oct 30. The draft already contains the PointPillars baseline, the stratified deletions, the controls, and this comparison. Pedestrian and Cyclist remain for the week of Nov 2. SECOND was not run.